

S. Andrew Lanham

Contact: andrewlanham55@gmail.com <https://scholar.google.com/citations?user=W6Gd4ZUAAAAJ>

RESEARCH STATEMENT

I have a multi-disciplinary background focusing on statistical inference and robust system design, spanning quantum information, analog computing, and signal processing. Current interest is in robust statistical characterization of noisy physical systems.

EDUCATION

The University of Texas at Austin, Austin, TX

Master of Science, Electrical Engineering, August 2019

- Thesis: Quantum Information Processing Approaches in Classical Systems (Advisors: Robert W. Heath Jr., Brian La Cour)

Bachelor of Science, Electrical Engineering, May 2017

EXPERIENCE

Co-Founder and Product Design

Austin, TX

Hardware Startup

January 2025 - Present

Design and distribution of complete embedded hardware and mechanical product in the area of food and beverage distribution, Embedded development, design, prototyping, strategic validation. Start-up environment.

- **Hardware Engineering:** PCB and embedded design prototyping and deployment, basic mechanical prototyping and deployment

Engineering Scientist

Applied Research Laboratories, UT Austin

Center for Quantum Research

August 2019 - January 2025

Led research and development at the intersection of quantum information science, advanced signal processing, and hardware systems. My work included architecting novel algorithms and benchmarking analog and digital computing systems against quantum devices.

- **Statistical Modeling:** Quantum-classical boundary, Gaussian statistics and post-selection. Simulation and analysis software development in Python
- **Statistical Modeling:** Benchmarking Analog Computing Systems. Modeling noise, detection strategies. Matlab, Python, and C/C++ software library design for signal processing and statistical benchmarking
- **Wireless Communication and Signal Processing:** New algorithms for wireless communication in the area of space-time block coding. Algorithm design and validation in Matlab, Python
- **Embedded Implementation:** Embedded system and PCB design using STM32F4 family

Graduate Research Assistant

Applied Research Laboratories, UT Austin

Center for Quantum Research

May 2017 - August 2019

Pioneered the experimental application of quantum error correction (QEC) to classical analog computing systems

Engineer, Systems Architecture

Tacen, Inc.

November 2017 - January 2019

Authored the formal technical documentation and system specifications for a high-performance, non-custodial digital asset exchange by communicating with project engineers. I was responsible for mathematically modeling and documenting the core logic to guide external engineering and stakeholder validation.

- **System Design and Documentation:** Authored end-to-end technical spec for a decentralized exchange backend and asset custody system
- **System Design and Documentation:** Interfaced with engineering teams to define and document modular interfaces for DEX backend components

Firmware Engineer and Product Design

Bad Hair Modular

November 2016 - September 2017

Designed and implemented the complete embedded firmware for a multi-channel digital/analog modular synthesizer module using a PIC32 microcontroller. My responsibilities included architecting the multi-mode user interface, managing real-time digital I/O, and controlling the I2C DAC for the analog output stages.

- **Embedded Design:** Designed PIC32-based firmware controlling a modular synthesizer's multi-mode envelope generation

PATENTS

Space-time encoding in wireless communication systems." U.S. Patent No. 11,218,195. 4 Jan. 2022.

JOURNAL PUBLICATIONS

Nandyala, Mahesh and **S.A. Lanham** and Jha, Prashant K. and Wu, Chengyue and Hazle, John D. and Yankeelov, Thomas E. and Stafford, R. Jason and El-Gendy, Ahmed A. and Fuentes, David. "An information-theoretic framework for optimal experimental design in magnetic nanoparticle hyperthermia." Applied Mathematical Modelling 116861. 2026

T.C. Cuvelier, **S.A. Lanham**, B.R. La Cour, and R.W. Heath Jr. "Quantum Codes for Classical Communication: A Space-Time Block Code from Quantum Error Correction." IEEE Open Journal of Communication. 2021

(*archival*) Lanham, S. Andrew. "Quantum-Inspired Approximations to Constraint Satisfaction Problems." arXiv preprint arXiv:2212.04016. 2022

CONFERENCE PUBLICATIONS

S.A. Lanham, and Brian R. La Cour. "Bell State Statistics with Post-Selection Loophole." Quantum 2.0. Optica Publishing Group, 2024.

S.A. Lanham, and Brian R. La Cour. "Quantum Resource States from Post-Selected Classical Mixed States." 2024 IEEE International Conference on Quantum Computing and Engineering (QCE). Vol. 1. IEEE, 2024.

S. Xie, M. Yang, **S.A. Lanham**, M. Wang, Y. Wang, S. Oruganti, J.R. Kulkarni. "Snap-SAT: A One-Shot Energy-Performance-Aware All-Digital Compute-in-Memory Solver for Large-Scale Hard Boolean Satisfiability Problems." ISSCC23. IEEE 2023

S.A. Lanham, E. Bradley, and B.R. La Cour. "Generalized Noncoherent Space-Time Block Codes from Quantum Error Correction." 2022 IEEE Military Communications Conference. IEEE 2022.

S.A. Lanham, and B.R. La Cour. "A Quantum-Inspired Classical Solver for Boolean k -Satisfiability Problems." 2021 IEEE International Conference on Quantum Computing and Engineering (QCE). IEEE 2021.

S.A. Lanham, and B.R. La Cour. "Detection-Based Measurements for Quantum Emulation Devices." 2020 IEEE International Conference on Quantum Computing and Engineering (QCE). IEEE, 2020.

C. Ostrove, B.R. La Cour, **S.A. Lanham**, and Granville Ott "Improving Performance of an Analog Electronic Device Using Quantum Error Correction," Journal of Physics Communications, accepted July 2019.

S.A. Lanham, T.C. Cuvelier, C. Ostrove, B.R. La Cour, G. Ott and R.W. Heath Jr. "A Non-coherent Space-Time Code from Quantum Error Correction," 2019 53rd Annual Conference on Information Sciences and Systems (CISS), Baltimore, MD, USA, March 2019, pp. 1-6.

B.R. La Cour, **S.A. Lanham** and C. Ostrove, "Parallel Quantum Computing Emulation," 2018 IEEE International Conference on Rebooting Computing (ICRC), McLean, VA, USA, 2018, pp.

1-9.

B.R. La Cour, G.E. Ott, and **S.A. Lanham**, “Using Quantum Emulation for Advanced Computation,” 2017 IEEE Custom Integrated Circuits Conference (CICC), Austin, TX, USA.